Do Now · retrieval

Paper 2 extended response challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a precondition?

Q2

What is a loop invariant?

Q3

What does O(1) auxiliary space mean?

Q4

For [1,1,2,3,3], count globally distinct values and consecutive runs.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Step 1 — State the sorted-input precondition and any edge cases.

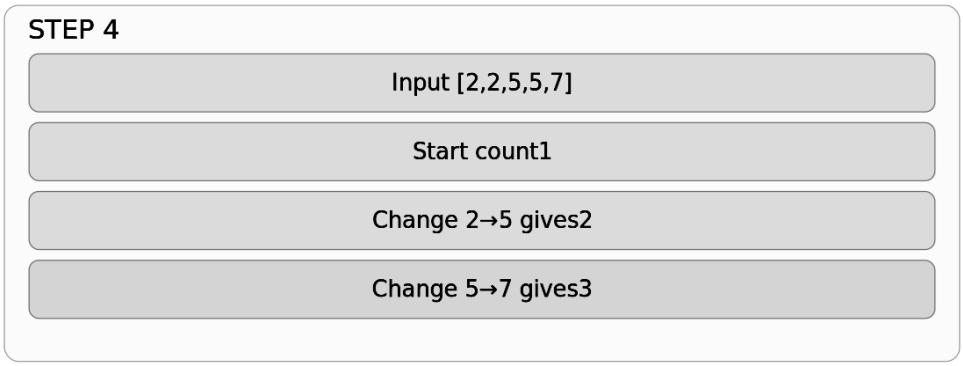
Core idea 2

Step 2 — Trace changing values with a loop invariant and correct bounds.

Core idea 3

Step 3 — Explain why the selected algorithm is efficient and when another is preferable.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

precondition — Required assumption about input.

invariant — Property maintained by a loop.

auxiliary space — Memory used beyond input.

hash set — Structure for expected efficient membership tests.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

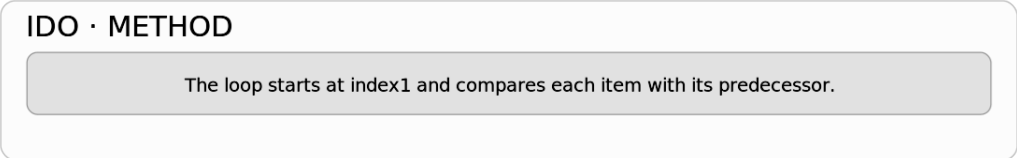
Problem

Trace UniqueCount for [1,1,2,3,3].

Follow the worked method

1. Explain the worked method — The loop starts at index1 and compares each item with its predecessor.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the worked method
The comparison counts new groups, not every repeated value.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write pseudocode to count distinct values in a sorted array, including empty input. [8]

Question 2

Trace [2,2,5,5,6,9,9] and give time/space complexities. [9]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate adjacent counting versus a hash set when the data is unsorted and contains a million elements. [9]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An adjacent comparison algorithm for sorted input works identically on unsorted input.

Use key vocabulary from memory

1. Explain precondition in your own words.

2. Explain invariant in your own words.

3. Explain auxiliary space in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.